DNS Resolver Exposure Assessment Report

Team A3-PS012-TC045 · AIORI Hackathon 3 · PS-012 · ResolverLens 0.2.0 · run **chal-a** · started 2026-10-28T08:53:21+00:00 · 6 authorised lab servers · 24 probes in 27.2 s

1. Summary

> **2 of 6** servers provide recursion to the external client: c04 (dnsmasq), c05 (unbound). Every decisive verdict matches the seeded ground truth. Accuracy 100.0% (6/6), accuracy on decided cases 100.0%, coverage 50.0% (3 of 6 servers decided; 3 correctly reported as Inconclusive). On the question "does it provide recursion?", a single RA-flag check gets 3/6 right and ResolverLens 6/6 (section 10).

Recursive (2) Authoritative-only (0) Refused (1) Inconclusive (3)

How to read the labels. **Recursive**: the server looked up a name for the external client (proved by our own canary server's log). **Authoritative-only**: it answers its own zone but does not recurse. **Refused**: it denies the external client. **Inconclusive**: the evidence is missing or contradictory, and the reason is recorded. Verdicts follow the rules on the method page (section 9).

2. Findings per server

Server	Address	Software	Verdict	Reason	Canary evidence (who asked the canary)	Ground truth (lab only)	Match
c01	10.10.0.41	dnsmasq	**Inconclusive**	CONFLICT	-	Inconclusive	**yes**
c02	10.10.0.42	dnsmasq	**Inconclusive**	CONFLICT	-	Inconclusive	**yes**
c03	10.10.0.43	bind	**Refused**	-	-	Refused	**yes**
c04	10.10.0.44	dnsmasq	**Recursive**	-	10.10.0.11	Recursive	**yes**
c05	10.10.0.45	unbound	**Recursive**	-	10.10.0.45	Recursive	**yes**
c06	10.10.0.46	bind	**Inconclusive**	CONFLICT	-	Inconclusive	**yes**

The ground-truth column exists only because this is a seeded lab. A real assessment has no such column.

3. Evaluation against the seeded ground truth

Rows are the expected labels, columns are what ResolverLens reported. Green cells are correct, red cells are errors.

Expected \ Reported	Recursive	Authoritative-only	Refused	Inconclusive	Total
Recursive	2	0	0	0	2
Authoritative-only	0	0	0	0	0
Refused	0	0	1	0	1
Inconclusive	0	0	0	3	3

Class	Precision	Recall	F1	Expected	Reported
Recursive	1.00	1.00	1.00	2	2
Authoritative-only	n/a	n/a	n/a	0	0
Refused	1.00	1.00	1.00	1	1
Inconclusive	1.00	1.00	1.00	3	3

Measure	Value	Meaning
Accuracy	100.0%	Verdicts equal to the expected label, over all servers
Accuracy on decided cases	100.0%	Correct verdicts among servers where the tool gave a decisive label

Coverage	50.0%	Share of servers the tool decided instead of reporting Inconclusive
Wrong decisions	0	Decisive verdicts that disagree with the expected label (the costly error)
False exposure / missed exposure	0 / 0	Reported Recursive when it should not be / failed to report Recursive when it should
Correct abstentions	3	Inconclusive reported for servers seeded as unusable

4. False and inconclusive result analysis

4.1 False results

> No false results in this run: all 6 verdicts equal their expected labels. Each false result would be listed here with a likely cause (ACL, caching, forwarder chain, packet loss, rate limiting, upstream failure).

4.2 Inconclusive results

Server	Software	Reason	Likely cause	Why the tool did not guess	Seeded as (lab only)	Expected?
c01	dnsmasq	CONFLICT	mixed evidence	The attempts disagreed or the flags, answer and canary log contradicted each other (caching, a forwarder chain or loss).	liar local answer	**yes**
c02	dnsmasq	CONFLICT	mixed evidence	The attempts disagreed or the flags, answer and canary log contradicted each other (caching, a forwarder chain or loss).	wildcard hijacker	**yes**
c06	bind	CONFLICT	mixed evidence	The attempts disagreed or the flags, answer and canary log contradicted each other (caching, a forwarder chain or loss).	sinkhole rewriter	**yes**

5. Exposure and remediation

RFC 5358 asks operators to offer recursion only to their intended clients. The following servers answered an external client:

Server	Software	Suggested change (check the syntax for your version)
c04 (10.10.0.44)	dnsmasq	`local-service (or interface=<internal interface> with bind-interfaces)` Its lookups came from 10.10.0.11: that upstream server needs the same restriction.
c05 (10.10.0.45)	unbound	`access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow`

6. Safety and compliance, measured from the evidence

Check	Observed	Limit or rule
Destinations contacted	6 (all on the allowlist: yes)	at most 20 authorised lab servers
Query types sent	A	type A only, never ANY
Peak rate, all targets together	3 queries in any second	5.0 per second plus burst
Peak rate, one target	3 queries in any second	1.0 per second plus burst
Largest reply / reply-to-query ratio	251 bytes / 6.28	1232 bytes / 10.0
How the run ended	completed	safety tripwire and stop file available

The tool has no spoofing option: every query leaves from the probe client's own address. The lab network has no Internet access.

7. Evidence per server

c01 · 10.10.0.41 · dnsmasq · Inconclusive (CONFLICT)

the evidence does not agree: canary statuses ['NOERROR', 'NOERROR', 'NOERROR'] (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	TIMEOUT	0	0	-	-	-
canary 1: rl-3a9ced6077.canary.lab	1	NOERROR	1	1	192.0.2.53	-	1.9
canary 2: rl-fdfc6eb2c2.canary.lab	1	NOERROR	1	1	192.0.2.53	-	0.8
canary 3: rl-abe045c986.canary.lab	1	NOERROR	1	1	192.0.2.53	-	0.8

c02 · 10.10.0.42 · dnsmasq · Inconclusive (CONFLICT)

the evidence does not agree: canary statuses ['NOERROR', 'NOERROR', 'NOERROR'] (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	1	192.0.2.53	-	1.2
canary 1: rl-6bb579b60f.canary.lab	1	NOERROR	1	1	192.0.2.53	-	0.8
canary 2: rl-1f3b702c12.canary.lab	1	NOERROR	1	1	192.0.2.53	-	0.8
canary 3: rl-ad3e58edc4.canary.lab	1	NOERROR	1	1	192.0.2.53	-	0.8

c03 · 10.10.0.43 · bind · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	REFUSED	0	0	-	-	1.0
canary 1: rl-105893efcd.canary.lab	1	REFUSED	0	0	-	-	0.7
canary 2: rl-d7a639ddf2.canary.lab	1	REFUSED	0	0	-	-	1.6
canary 3: rl-bd36f0b79c.canary.lab	1	REFUSED	0	0	-	-	0.9

c04 · 10.10.0.44 · dnsmasq · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	4.8
canary 1: rl-b31be63324.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	2.0
canary 2: rl-c38475f1d1.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	2.2
canary 3: rl-703ba33455.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	2.2

c05 · 10.10.0.45 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NXDOMAIN	1	1	-	-	201.6
canary 1: rl-f3037da769.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.45	402.0
canary 2: rl-add5960fca.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.45	401.6
canary 3: rl-4a8ad5ccd1.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.45	401.5

c06 · 10.10.0.46 · bind · Inconclusive (CONFLICT)

the evidence does not agree: canary statuses ['NOERROR', 'NOERROR', 'NOERROR'] (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	2.8
canary 1: rl-da39710c4c.canary.lab	1	NOERROR	1	0	192.0.2.53	-	1.2
canary 2: rl-40b1ec8041.canary.lab	1	NOERROR	1	0	192.0.2.53	-	1.6
canary 3: rl-6df12bfb00.canary.lab	1	NOERROR	1	0	192.0.2.53	-	1.2

8. Reproducibility

Item	Value
Run, seed, tool version	chal-a, 1337, 0.2.0
Config digest (settings and targets file)	`dbd4f1b9209fe8c22f412f95f0e6d54283a1c25560a9df971d61348306658f9f`
Timeout, attempts, EDNS buffer	2.0 s, 3, 1232 bytes
Re-run	`make probe RUN=<name>` then `make report RUN=<name>`. The same seed and run id give the same canary names, and the verdicts are computed from evidence.jsonl alone.

9. Method: the four rules

Each server receives one baseline query (RD off, a name in its own zone) and 3 canary queries (RD on, a fresh unique name each time, in a zone served only by our canary server). The canary server logs every query it receives. Rules are checked in order and the first match decides:

Rule	Verdict	Condition
1	Recursive	Every canary attempt is NOERROR with RA set, the marker answer, and a hit in the canary log
2	Refused	The baseline and every canary attempt are REFUSED
3	Authoritative-only	The baseline is answered with AA; every canary attempt gets a reply with no answer, no RA and no canary hit
4	Inconclusive	Anything else, with a reason: TIMEOUT, SERVFAIL, TC_FAIL, CONFLICT, ERROR or NO_DATA

10. Comparison with single-flag checks

The same evidence judged by simpler rules, on one question: does the server provide recursion to the external client? A single-flag check reads only the first RD=1 reply. That is not enough when a server can set RA and return an answer without ever recursing (a sinkhole, a DNS firewall, a local-data override). ResolverLens also requires the canary server's log to show the lookup.

Method	What it reads	Correct	False exposure	Missed exposure	Servers called wrongly
ResolverLens	flags, answer and canary evidence over 3 attempts	6/6	0	0	none
RA-flag check	first RD=1 reply: NOERROR and RA set (what Nmap's dns-recursion script reads)	3/6	3	0	c01, c02, c06
RA + AA check	first RD=1 reply: NOERROR, RA set, AA clear and an answer	5/6	1	0	c06

Server	Seeded as (lab only)	Truly recursive?	ResolverLens	RA-flag check	RA + AA check
c01	liar local answer	no	**not recursive**	**recursive**	**not recursive**
c02	wildcard hijacker	no	**not recursive**	**recursive**	**not recursive**
c06	sinkhole rewriter	no	**not recursive**	**recursive**	**recursive**